

High Performance Waveguide GeSi Avalanche Photodiode Fully-integrated on a 220 nm Silicon Photonics Platform

Jiliang Wu⁽¹⁾, Fan Qi⁽¹⁾, Pengfei Cai⁽¹⁾, Bin Shi⁽¹⁾, Chingyin Hong⁽¹⁾, Tehuang Chiu⁽¹⁾, Guanghui Hou⁽¹⁾, Jiaxing Zhao⁽¹⁾, Ming Sun⁽¹⁾, Zengwen He⁽¹⁾, Dong Pan⁽¹⁾

SiFotonics Technologies Co., Ltd., Suite 101, 10 New England Business Center Drive, Andover, MA 01810, USA

dp7u@sifotonics.com

Abstract We demonstrate a 100Gbps Ge/Si waveguide APD integrated with low loss SiN edge couplers and other components on a 220nm SOI platform, achieving ultra-low dark currents and record sensitivity. Our results offer a high-density integration path for APDs in high-link budget and low-power dissipation scenarios. ©2026 The Author(s)

Introduction

In recent years, waveguide-integrated Germanium-on-Silicon (Ge/Si) photodetectors (PDs) have become indispensable building blocks in integrated photonics, enabling high speed optical data transmission. To further scale the transmission rate and enhance the link budget, Ge/Si waveguide avalanche photodiodes (WGAPDs) on the silicon photonics platform have been extensively studied for their internal gain, low excess noise factor, and seamless CMOS compatibility [1-2].

Waveguide APD are conventionally designed into lateral or vertical junction configurations. Lateral junction APDs utilize the 220 nm Si device layer as the multiplication region, offering fabrication simplicity and design flexibility without requiring additional intrinsic silicon epitaxy [3]. Although lateral APDs can deliver GBPs surpassing 1000 GHz [4-6], a high dark current remains a persistent challenge that limits robust transmission in high temperature operating conditions of higher than 85 °C. On the other hand, vertical junction APDs enable precise electric field control inside the Ge on Si structure thus a low dark current levels of less than 1 uA [7-8]. The primary challenge in fabricating vertical WGAPDs on a 220 nm SOI platform is the requirement for an additional intrinsic silicon epitaxy step to serve as an avalanche multiplication layer. 50 Gbps-capable vertical WGAPDs have been demonstrated with dark currents below 150 nA on a 300 mm silicon photonics platform [9]. However, as the industry transitions toward 1.6 T and 3.2 T optical interconnects, there is an urgent demand for monolithically integrated APDs that can support significantly higher data rates while maintaining high sensitivity. On the other hand, passive devices are also key elements in silicon photonics, such as edge coupler and polarization splitting rotator (PSR). However, the integration of silicon nitride (SiN) waveguide layers has not been reported in vertical WGAPDs where extra silicon epitaxy needed.

In this paper, we report a 100 Gbps Ge/Si WGAPD monolithically integrated on a standard 220 nm SOI platform with one silicon nitride waveguide layer. High performance edge coupler and APD are realized at the same time. Also, to address practical application requirements, an on-chip Bias-T and a PSR are integrated with the WGAPD. This fully integrated architecture enables a record high sensitivity 100 Gbps optical receiver with significantly reduced polarization-dependent loss (PDL), providing a robust solution for next-generation high-speed optical interconnections.

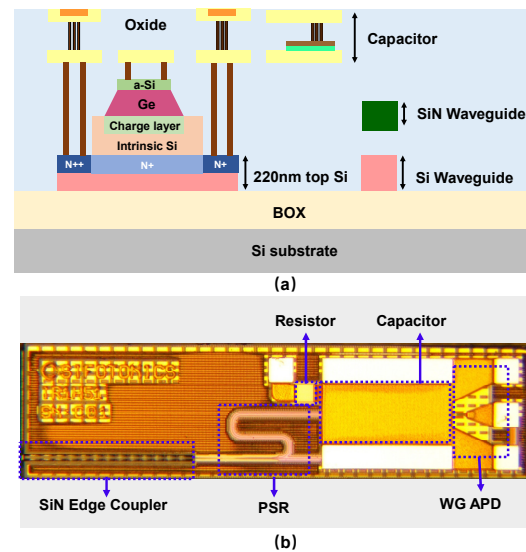


Fig. 1: (a) cross-section view and (b) microscopic image of the integrated Ge/Si waveguide avalanche photodiode.

Device structure and Fabrication

Figure 1 presents the cross-section view and the microscopic image of the Ge/Si WGAPD. The monolithically integrated APD chip was fabricated on a 200 mm wafer with 220 nm thick top silicon and 3 μm buried oxide (BOX). The APD adapted a vertical Separate Absorption Charge Multiplication (VSACM) architecture. High quality silicon and germanium layers were grown se-

quentially by a carefully tuned recipe, which guarantees a low threading dislocation density at Ge/Si interface. To simultaneously achieve high bandwidth and low dark current, the electrical field distribution in the multiplication and absorption layer was precisely manipulated.

Besides the waveguide APD core device, an integrated Bias-T was also constructed for a low-complexity package, as shown in Fig.1 (b). The integrated Bias-T consists of an on-chip resistor for DC biasing and a series capacitor for RF signal coupling. The resistor was defined by ion implantation into the top silicon layer of the SOI wafer and the capacitor was integrated between the Metal-1 and Metal-2 layers. Furthermore, to minimize the PDL between TE and TM modes, a polarization splitting rotator (PSR) and a silicon SiN edge coupler are monolithically integrated on the waveguide APD chip.

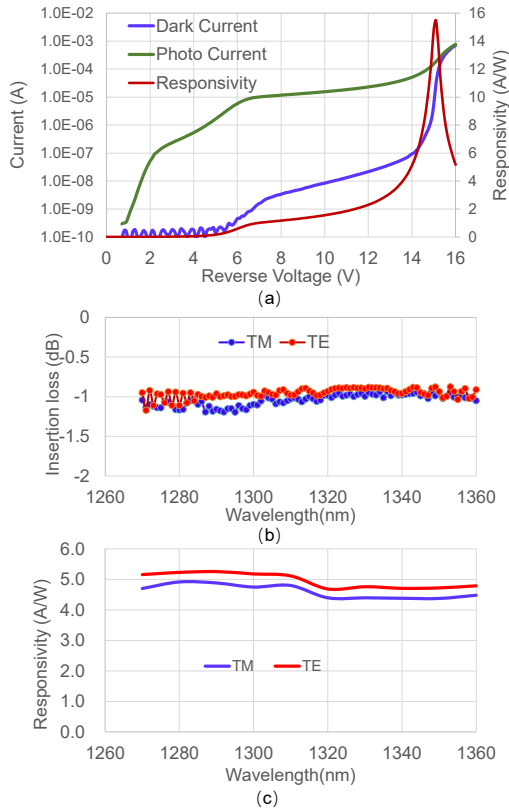


Fig. 2: (a) I-V and responsivity of the WGAPD (b) coupling loss of the SiN edge coupler with single mode fiber and (c) responsivity versus wavelength measured via single mode fiber of the WGAPD.

Device characteristics

Figure 2(a) shows the dark current, photo current and responsivity characteristics of the waveguide APD at room temperature. The measured photo-current of the waveguide APD includes the coupling loss from the fibre to the waveguide. The breakdown voltage (V_{br}), defined as the applied voltage when the dark current equals $10 \mu A$, is close to 15 V. When the APD is biased at 14 V

($\sim 0.95 V_{br}$), the dark current is only 80 nA and responsivity is 5 A/W at 1310nm with an input power of -20 dBm. The primary responsivity of the WGAPD is 0.78 A/W at 1310 nm extracted via a waveguide PD structure with the same Ge/Si thicknesses considering a 1.0 dB coupling loss, as shown in Fig.2 (b). Fig. 2(c) shows WGAPD's O-band responsivity for both TE and TM modes at 14 V. It shows a low PDL of less than 0.4 dB. Considering that the SiN edge coupler has a PDL of 0.2 dB, the WGPAD's PDL is only 0.2 dB thanks to the on-chip PSR. Moreover, the WGAPD has a good responsivity uniformity of more than 4.5 A/W across the O band, which benefits CWDM4 400/800GbE transmission.

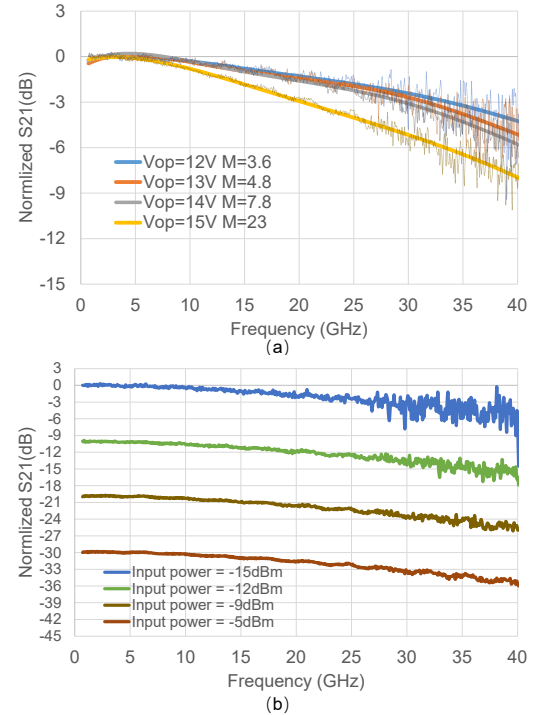


Fig. 3: normalized S_{21} of the WGAPD under (a) different bias voltage and (b) different optical input power.

The small signal RF performance of the WGAPD has been characterized via 67 GHz Network Analyser at 1310 nm. Fig.3(a) shows the normalized S_{21} (opto-electric frequency response) curves under different bias voltage. The raw S_{21} data are fitted to calculate the 3-dB bandwidth. At 14V, the extracted bandwidth is 30 GHz with gain $M=7.8$. Besides, the S_{21} curves of the WGAPD show consistent RF performance from -15 dBm to -5 dBm in Fig.3 (b), indicating a good linearity for receiver. Furthermore, RC limited bandwidth is calculated to be 65 GHz considering the WGAPD's junction capacitance of 20 fF. Thus, the high speed performance of the WGAPD is constrained by the carrier transit-time and avalanche build-up time. Further bandwidth enhancement can be achieved by optimizing the Ge/Si film stack.

WGAPD receiver performance

To further test the high-speed transmission performance of the WGAPD, we packaged the APD on an evaluation board with a 50-Gb/s transimpedance amplifier (TIA), as shown in Fig 4 (a). Owing to the monolithic integration of the bias-T on the WGAPD, off-chip capacitors and resistors is eliminated. This not only simplifies the packaging but also minimizes the parasitic effects associated with wire-bonding. The WGAPD receiver has a 3-dB bandwidth of ~ 30 GHz. Moreover, no obvious peaking is found on the S₂₁ curve, indicating a low parasitic inductance.

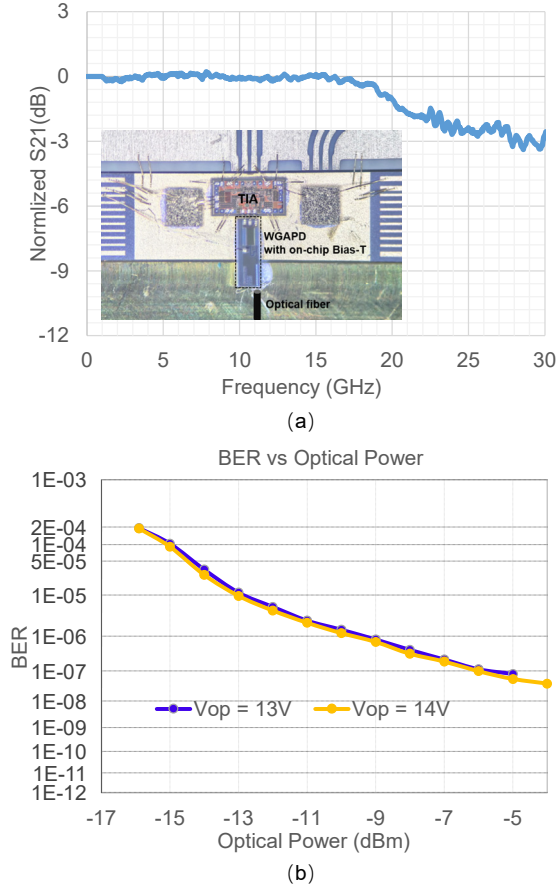


Fig. 4: (a) RF response and (b) BER results of the WGAPD receiver

A commercial DR1 transceiver ($\lambda = 1310$ nm) serves as the 100 Gbps PAM4 optical source, exhibiting a TDECQ of 2 dB and an outer extinction ratio (ER) of 5.5 dB. Fig. 3(b) illustrates the measured back-to-back (B2B) 100 Gbps PAM4 sensitivity curves. At a bias voltage of 13 V, a receiver sensitivity of -16.0 dBm was achieved at a BER of $2\text{E-}4$ excluding optical insertion loss. Additionally, the measured sensitivity is nearly identical for both TE and TM modes, demonstrating the effective polarization-insensitive operation of the WGAPD. This performance is comparable to the state-of-the-art 100 Gbps APDs [9], demonstrating the high efficiency of our waveguide-integrated design.

Table 1 summarizes the performance of reported WGAPD based on a 220 nm SOI silicon photonics platform. Our WGAPD shows a record 100 Gbps sensitivity and a lowest dark current of less than 100 nA, which supports 85°C high temperature operation without sensitivity penalty. Remarkably, we first demonstrate a fully integrated WGAPD architecture combined with on-chip Bias-T, PSR and SiN waveguide. Our results expand the functionalities of the SiN/SOI photonics platform for high density and low-power dissipation optical interconnects.

Conclusions

In this paper, we demonstrate a 100 Gbps Ge/Si waveguide avalanche photodetector monolithically integrated on a standard 220 nm SOI platform. The WGAPD has a ultra-low dark current of 80 nA at $M=7.8$ with a 3-dB BW larger than 30GHz. Low loss SiN waveguide, on-chip Bias-T and PSR are also integrated in the WGAPD chip, achieving a PDL of less than 0.2 dB and significant packaging advantages. This fully integrated WGAPD constructs a 100 Gbps optical receiver with record sensitivity up to -16 dBm. Our monolithic WGAPD architecture offers a high-density integration path for large-scale and low-power dissipation silicon photonic chips.

Tab. 1: performance of Ge/Si WGAPD on 220nm SOI platform

Type	λ (nm)	Resp. (A/W)	Gain	BW (GHz)	Id (μA)	Integrated building block	KP4 FEC Sensitivity(dBm)	Ref.
Lateral	1310	0.93	12.8	48	100	NA	-11.3 (100 Gbps PAM4)	[4]
Lateral	1550	0.87	19.5	53	12	NA	-13.0 (100 Gbps PAM4)	[5]
Vertical	1310	0.5	4	35	0.15	NA	Eye diagram with 6.1 dB SNR (50 Gbps NRZ)	[8]
Vertical	1550	1.1	20	25	~ 1.0	NA	-13.0 (64 Gbps PAM4)	[10]
Vertical	1550	1.12	12	25	~ 0.5	NA	-15.7 (32 Gbps NRZ)	[11]
Vertical	1310	0.78	7.8	30	0.08	Bias-T, PSR SiN waveguide	-16.0 (100 Gbps PAM4)	This work

References

- [1] S. A. Srinivasan, M. Berciano, P. De Heyn, S. Lardenois, M. Pantouvaki and J. Van Campenhout, "27 GHz Silicon-Contacted Waveguide-Coupled Ge/Si Avalanche Photodiode," *Journal of Lightwave Technology*, vol. 38, no. 11, pp. 3044-3050, 2020, DOI: [10.1109/JLT.2020.2986923](https://doi.org/10.1109/JLT.2020.2986923).
- [2] S. Yanikgonul, X. Luo and G. -Q. Lo, "High-Speed Waveguide-Integrated Avalanche Photodiodes for Near-Infrared Wavelengths on SiN-on-SOI Photonic Platform," *Journal of Lightwave Technology*, vol. 41, no. 6, pp. 1811-1819, 2023, DOI: [10.1109/JLT.2022.3225464](https://doi.org/10.1109/JLT.2022.3225464).
- [3] Ahmed Shahin, Mohamed A. Swillam, and Mohamed S. El-Nabi, "Design of a High-Speed Waveguide Ge/Si SACM Avalanche Photodiode," *Journal of Lightwave Technology*, vol. 43, no. 19, pp. 9325–9332, 2025, DOI: [10.1109/JLT.2025.3594842](https://doi.org/10.1109/JLT.2025.3594842).
- [4] Yuluan Xiang, Hengzhen Cao, Chaoyue Liu, Jingshu Guo, and Daoxin Dai, "High-speed waveguide Ge/Si avalanche photodiode with a gain-bandwidth product of 615 GHz," *Optica*, vol. 9, no. 7, pp.762-769, 2022, DOI: [10.1364/OPTICA.462609](https://doi.org/10.1364/OPTICA.462609).
- [5] Yang Shi, Xiang Li, Guanyu Chen, Mingjie Zou, Hongjun Cai, Yu Yu, and Xinliang Zhang, "Avalanche photodiode with ultrahigh gain–bandwidth product of 1,033 GHz," *Nature Photonics*, vol. 18, no. 9, pp. 933–939, 2024, DOI: [10.1038/s41566-024-01421-2](https://doi.org/10.1038/s41566-024-01421-2).
- [6] Yang Shi, Mengting Zou, Zhicheng Li, Jinyu Zhang, Shuhan Liu, Jinsong Xia, Yu Yu, and Xinliang Zhang, "Ultra-fast avalanche photodiode exceeding 100 GHz bandwidth," *Nature Communications*, vol. 16, no. 1, art. no. 11058, 2025, DOI: [10.1038/s41467-025-66047-6](https://doi.org/10.1038/s41467-025-66047-6).
- [7] Mengyue Huang, Zeyu Cui, Yuntao Hu, Zidong Feng, Jiesi Chen, Zanyun Zhang, and Shao-Bing Liu, "Recess-type waveguide integrated germanium on silicon avalanche photodiode," 2021 Optical Fiber Communications Conference and Exhibition (OFC), San Francisco, CA, USA, 2021, pp. 1–3. DOI: [10.1364/OFC.2021.Tu5H.2](https://doi.org/10.1364/OFC.2021.Tu5H.2).
- [8] A. Tsiara, G. Lepage, P. Verheyen, J. Van Campenhout, and M. Pantouvaki, "50 Gbps vertical separate absorption charge multiplication Ge/Si avalanche waveguide photodetectors integrated in a 300-mm Si photonics platform," 49th European Conference on Optical Communications (ECOC 2023), Glasgow, UK, 2023, pp. 1170–1173. DOI: [10.1049/icp.2023.2486](https://doi.org/10.1049/icp.2023.2486).
- [9] Bin Shi, Yanhui Duan, Fan Qi, Pengfei Cai, Xueping Chen, Zhenhua Wei, Guanghui Hou, Tzungli Su, Su Li, Wang Chen, Chingyin Hong, and Dong Pan, "Record Performance of 53GB Ge/Si Avalanche Photodiode ROSA," 2021 Optical Fiber Communications Conference and Exhibition (OFC), San Francisco, CA, USA, 2021, pp. 1–3. DOI: [10.1364/OFC.2021.M5A.4](https://doi.org/10.1364/OFC.2021.M5A.4).
- [10] Binhao Wang, Zhihong Huang, Yuan Yuan, Di Liang, Xiaoge Zeng, Marco Fiorentino, and Raymond G. Beausoleil, "64 Gb/s low-voltage waveguide SiGe avalanche photodiodes with distributed Bragg reflectors" *Photonics Research*, vol. 8, no. 7, pp.1118-1123, 2020, DOI: [10.1364/PRJ.390339](https://doi.org/10.1364/PRJ.390339).
- [11] Yuan Yuan, Zhihong Huang, Xiaoge Zeng, Di Liang, Wayne V. Sorin, Marco Fiorentino, and Raymond G. Beausoleil. "High Responsivity Si-Ge Waveguide Avalanche Photodiodes Enhanced by Loop Reflector" *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 28, no. 2, pp.1-8, 2022, DOI: [10.1109/JSTQE.2021.3087416](https://doi.org/10.1109/JSTQE.2021.3087416).